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Hybrid deep learning architecture for skin disease classification

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ABSTRACT

Early detection and accurate classification of skin diseases is clinically important and required for proper diagnosis and treatment because delayed diagnosis can lead to inadequate treatment and increased patient risk, especially when visually similar conditions require different management approaches. However, automated skin disease detection remains challenging due to subtle inter-class differences and the need to capture both individual lesion textures and patterns. In this paper, we propose a lightweight hybrid model to detect and classify skin diseases accurately with less computational complexity. The proposed hybrid deep learning architecture combines complementary representations from two models, MaxViT and MobileNet. MaxViT uses a multi-axis self-attention mechanism that decomposes global attention into block-wise and grid-wise operations, reducing the quadratic self-attention complexity, while MobileNet effectively extracts spatial features using depth-wise separable convolutions with minimal parameters. We evaluate the performance of the proposed hybrid method on two different Datasets, including eight classes of skin disease images. The proposed hybrid model obtains an accuracy of 99.57% and 98.67% using Dataset 1 and Dataset 2, respectively. The results indicate that by combining complementary features from two models can ensure better screening accuracy while maintaining less computational complexity and processing time, which is required for real-time clinical workflow decision support.

1. Introduction

The skin is the largest organ, both externally and in general. Numerous patients have reported diverse patterns of skin disorders throughout multiple nations [1]. Skin diseases affect Bangladeshis from 11.16 to 63 percent of the population as a whole [2]. Skin disease varies among individuals of different ages of the same region. However, viral, bacterial, fungal, and parasite skin infections are highly prevalent causing very little death, but have a chronic course and cause agony [3]. Skin diseases caused by fungi, such as athlete's foot, nail fungus, and ringworm; bacteria, such as cellulitis and impetigo; viruses, such as chickenpox and shingles; and parasitic infections, such as cutaneous larva migrans, are common in Bangladesh due to factors like overcrowding, poor sanitation, socioeconomic conditions, excessive use of chemicals and cosmetics, environmental pollution, and irregular weather patterns that result in temperature increases and precipitation. In a 2018 study conducted at Birdem General Hospital in Dhaka, Bangladesh [1], it was discovered that 26,980 of the 2,62,870 OPD-visited patients had a skin condition, with parasitic, fungal, bacterial, and viral infections accounting for roughly 33%, 28%, 20%, and 9.56% of cases, respectively. Therefore, a greater attention to skin diseases is necessary to improve the situation.

The importance of computer-aided methods for diagnosing skin disorders is growing with time. A large range of machine learning and deep learning techniques are available to classify skin diseases which includes several preprocessing steps and feature extraction methods. In [4], the author introduces image preprocessing techniques with feature extraction and the Support Vector Machine (SVM) machine learning model to classify three different types of skin diseases and demonstrates promising results on a small dataset. The authors have applied a similar approach in [5] to predict skin infections and to compare it with neural network performance. However, the extraordinary accuracy reported in [4] and [5] is limited to a small dataset with no external validation, raising concerns about overfitting without cross-validation and dataset bias. These factors significantly constrain confidence in the claimed performance and generalizability. In [6] and [7], a comprehensive survey and performance comparison of different machine learning classifiers and transfer learning approaches like SVM, VGGNet, and Inception-ResNet-V2 applied on different sizes of malignant melanoma skin cancer datasets have been presented and it is reported that SVM performs the best, achieving 60% to 97% accuracy based on the size of the dataset. The survey in [7] shows a comparative analysis of preprocessing, segmentation, feature extraction

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and classification for skin cancer detection using the extracted features. However, feature extraction before applying machine learning requires in-depth exploratory data analysis to minimize the dimension, and a high level of professional medical understanding. Further, the findings can finally be generated after a difficult parameter adjustment, which requires time and effort.

Deep learning algorithms dramatically improve the effectiveness and gained popularity in fine-grained medical imaging classification problems [8,9]. DL models require large number of nodes and employ a deep complex architecture, compared to machine learning methods. In [10], it is shown that transfer learning methods, that use pre-trained deep learning classification models, can be applied to obtain a superior result and decrease overfitting using a small dataset. The authors in [11] use different pre-trained convolutional neural network (CNN) algorithms as ResNet50, InceptionV3, DenseNet121, Xception and Inception-ResNetV2 to classify six common skin diseases using 2656 face image datasets including 388 test datasets showing the best possible recalls of 92.9%, 89.2% and 84.3% to classify three diseases and the mean recall and precision of 77.0% and 70.8%, respectively. In [12], the authors present two cross-domain skin disease recognition methods using a fully supervised classifier pre-trained on imagenet and adversarial domain adaptation technique showing improved recognition performance of about 92.3% on melanoma and cancer detection. The authors in [13] proposed a hybrid method to address the class imbalance issue of skin disease classification by combining the data-level approach of a balanced mini-batch logic followed by real-time image augmentation to overcome the poor learning of the minority classes and achieve a high accuracy of 89.97% using their proposed EfficientNetB4-CLF model. The authors in [14] propose a novel methodology by leveraging the deep tuning layers of the two distinct deep learning algorithms with a triple loss function to obtain an accuracy of 87.42% in skin disease classification. In [15], the authors develop a personalized, extendable, real-time skin disease detection system using self-learning, data width evolution, information entropy-based data set filter method, and three deep learning models; LeNet-5, AlexNet, and VGG16 demonstrating a reliable and effective system achieving the best accuracy of 84.6% on average using AlexNet. The authors in [16] introduce a genetic CNN algorithm called SEECNN using a straightforward encoding approach in which population members represent the parameters and weights of each layer addressing the large search areas and variable-length optimization issues demonstrating its efficacy in skin disease classification outperforming the state-of-the-art approaches. The authors in [17] discuss Generative Adversarial Networks, Meta-Learning, Few-Shot classification, and 3D face modeling to improve computer-aided skin disease diagnosis on rosacea addressing the data scarcity problems. With the use of cutting-edge methods like CNN, the authors in [18] extract the skin disease image features and classify them with an accuracy of 87%.

However, deep learning models cannot perform the best without a significant amount of training data which might not be available in many real-world scenarios. According to [19], deep learning models tend to overfit and increase variance when trained on smaller datasets. Transfer learning models are deep learning models that have already been trained on small datasets to minimize variance and overfitting. An ensemble of classifiers is often far more accurate than the sum of its parts [20]. An ensemble deep learning approach for skin disease classification is proposed in [21] using Fuzzy C Means segmentation (FCM) on preprocessed images and then applying deep feature extraction using Resnet50, VGG16, and Deeplabv3. Then by applying feature transformation using Hybrid Squirrel Butterfly Search Optimization (HSBSO) and then applying Modified Long Short Term Memory (MLSTM) for classification, an improved accuracy in skin disorders classification is obtained. The authors in [22] introduce ConvNeXt-ST-AFF, which uses Attentional Feature Fusion (AFF) to combine local and global features of ConvNeXt and Swin Transformer networks. The hybrid model outperforms existing models showing the highest classification accuracy of 92.16%.

In this paper, we apply both deep learning and transfer learning models, and ensemble-based approaches to obtain an optimal result in the classification of eight skin diseases. In this paper, we used two open-source datasets of different volume; Skin Disease Dataset [23] and the ISIC 2019 Skin Lesion Images for classification dataset [24] obtained from Kaggle for classification, demonstrating better generalization capability for skin disease recognition. The contributions of the paper is summarized as in the following.

1. First, we applied data augmentation and data pre-processing methods.
2. Then, we employ numerous pre-trained transfer learning models based on deep learning, such as VGG16, ResNet101, MaxViT and MobileNet to address the overfitting problem and to compare the performance.
3. Then, we propose a hybrid model for skin disease classification that fused an advanced version of Vision Transformer, MaxViT, and a deep learning architecture, MobileNet.
4. Then, we also develop a convolutional neural network (CNN) architecture for skin disease classification.
5. We also apply ensemble-based approaches consisting of multiple algorithms to compare the performances with our proposed hybrid model.

It is observed that the ensemble-based approach significantly improves performance in terms of Accuracy, Precision, Recall, and F1 score compared to a single model, enhancing the detection of skin diseases. Using our proposed hybrid model, we achieved an improved multi-classification accuracy of 99.57% for the Skin-Disease-Dataset and 98.67% accuracy for ISIC 2019 Skin Lesion Images for Classification dataset.

The paper is organized to systematically present the proposed study. First, the methodology section describes the proposed approach, including data source, data preprocessing, feature extraction, and the machine learning, deep learning, transfer learning and proposed hybrid models employed. This is followed by the experimentation and results section, which outlines the experimental setup, evaluation metrics and the quantitative performance outcomes, while the discussion section analyzes the findings and compares them with existing works. Finally, the paper concludes by summarizing the main contributions and highlighting potential directions for future research.

2. Proposed methodology

The proposed skin disease classification system processes skin disease images to detect one of the eight diseases. The methodologies include data augmentation, pre-processing, and classification using hybrid architecture, customized convolutional neural networks (CNN), pre-trained deep learning models, and a proposed ensemble-based approach. The methodologies are explained in the following.

2.1. Data source

Our study uses two publicly available open-source datasets obtained from Kaggle: Dataset 1, the Skin Disease Dataset [23], and Dataset 2, the ISIC 2019 Skin Lesion Images for Classification [24]. The Skin Disease Dataset comprises 1159 labeled images representing eight distinct categories of skin conditions. These include bacterial infections (cellulitis and impetigo), fungal infections (athlete's foot, nail fungus, and ringworm), parasitic infections (cutaneous larva migrans), and viral skin infections (chickenpox and shingles).

The ISIC 2019 Skin Lesion Images for Classification dataset contains 25,331 dermoscopic images distributed in eight classes, namely Melanoma (MEL), Melanocytic Nevus (NV), Basal Cell Carcinoma (BCC),



Fig. 1. Dataset sample: (a) The Skin Disease Dataset (b) The ISIC 2019 Skin Lesion Images for Classification Dataset.

Table 1
Augmentation techniques used for Dataset 1: The Skin Disease Dataset and Dataset 2: The ISIC 2019 Skin Lesion Images for Classification dataset.

Dataset	Technique	Description
Dataset 1	Rotation	Randomly 10 degrees left or right
	Zoom	Factor between 1.1 and 1.6 of their original size
Dataset 2	Rotation	Two fixed angles of 45 and 90 degrees
	Zoom	Factor between 0.8 and 1.2 of their original size
	CLAHE	Use clip limit 1.5

Table 2
Training–validation split for Dataset 1 and Dataset 2.

Dataset	Class	Training	Validation	Total
Dataset 1	BA-Cellulitis	1000	34	1034
	BA-Impetigo	1000	20	1020
	FU-Athlete-foot	1000	32	1032
	FU-Nail-fungus	1000	33	1033
	FU-Ringworm	1000	23	1023
	PA-Cutaneous-Larva-Migrans	1000	25	1025
	VI-Chickenpox	1000	34	1034
	VI-Shingles	1000	33	1033
	Total	**8000**	**234**	**8234**
Dataset 2	Actinic Keratosis (AK)	3486	849	4335
	Basal Cell Carcinoma (BCC)	13 334	3281	16 615
	Benign Keratosis (BKL)	10 514	2606	13 120
	Dermatofibroma (DF)	955	240	1195
	Melanoma (MEL)	18 031	4579	22 610
	Melanocytic Nevus (NV)	51 467	12 908	64 375
	Squamous Cell Carcinoma (SCC)	2510	630	3140
	Vascular Lesion (VASC)	1027	238	1265
	Total	**101 324**	**25 331**	**126655**

Actinic Keratosis (AK), Benign Keratosis (BKL), Dermatofibroma (DF), Vascular Lesion (VASC), and Squamous Cell Carcinoma (SCC). Representative sample images from both datasets are illustrated in Figs. 1(a) and 1(b).

2.2. Data augmentation and pre-processing

The Skin Disease Dataset contains a relatively small number of samples, which is insufficient for training a robust and reliable model. The original training set includes 925 images across eight classes. To address this limitation, data augmentation techniques were applied to the training set. Specifically, images were randomly rotated by up to 10 degrees to the left or right and zoomed with a scaling factor ranging from 1.1 to 1.6 relative to their original size. Through these augmentations, the training set was expanded to approximately 8000 images across the eight classes.

Similarly, augmentation strategies has been also applied to the ISIC 2019 Skin Lesion Images for Classification dataset. Contrast Limited Adaptive Histogram Equalization (CLAHE) is used to enhance image contrast with a clip limit of 1.5. Additionally, images are zoomed using a scaling factor between 0.8 and 1.2 and rotated at two fixed angles: 45° and 90°. The augmentation techniques and the final number of images after augmentation for both datasets are reported in Tables 1 and 2, respectively.

All images from both datasets are converted to RGB format and resized to 224 × 224 pixels. After augmentation and resizing, the images are converted into tensors and normalized channel-wise using the ImageNet mean and standard deviation (μ = [0.485, 0.456, 0.406], σ = [0.229, 0.224, 0.225]). This preprocessing aligns the inputs with pretrained models, stabilizes the optimization process, and supports reliable transfer learning performance. For model evaluation, the datasets are split into training and validation sets using an 80–20 ratio.

2.3. Pre-trained deep learning models

Deep learning models can differentiate between several image classes with higher accuracy by distinguishing complex patterns in images. Additionally, the smaller datasets of this work suggest the usage of Transfer Learning in which previously trained Deep Learning models can be applied to a current dataset to generate predictions. In this paper, we have utilised various deep learning models, including VGG16, ResNet101, MobileNet, and MaxViT. The overview of the models is shown in Fig. 2 and is explained in the following.

2.3.1. VGG16

VGG16 is a deep neural network consists of 13 convolutional layers with 3 × 3 filters and “same” padding make up its 16 layers followed by maxpooling layers to minimize spatial dimensions. It also includes three dense layers and an output layer of 1000 nodes. The model can extract both high-level features and fine-grained information which is necessary for the detection and classification of skin illnesses as dermatological conditions. We customize the model to optimize the performance for skin disease classification.

First, the fully connected dense layers at the top of the model are removed. Next, the output layers are flattened and then, the fully connected layer is customized with two dense layers. Faster convergence and improved accuracy are guaranteed when the Adam optimizer for adaptive learning is combined with ReLU activation function for non-linearity and three classes are predicted at the output layer with the SoftMax activation function. Finally, VGG16 becomes incredibly effective due to the pre-trained weights from massive datasets like ImageNet.



Fig. 2. Overview of the pre-trained transfer learning models.

2.3.2. ResNet101

ResNet101 is a Deep convolutional neural networks frequently utilized for effective deep model training. A pre-trained ResNet101 based on ImageNet is optimized for the classification of skin diseases. To customize the model for classification, the ResNet101 foundation is frozen and only custom layers such as a global average pooling layer and fully connected layers with ReLU and softmax activations are trained. Using a batch size of 64 across 500 epochs, the Adam optimizer and categorical cross-entropy have been used.

2.3.3. MobileNetV3

MobileNetV3 is a lightweight convolutional neural network designed to achieve high accuracy while utilizing less computational resources and memory. It employs efficient building blocks, such as depthwise separable convolutions and inverted residual connections, allowing it to learn meaningful visual information with fewer parameters. MobileNetV3 also incorporates attention-like modules (such as squeeze-and-excite) and enhanced activation algorithms to help you focus on the most crucial aspects of an image.

2.3.4. MaxViT-Tiny

MaxViT-Tiny is a modern vision model that harnesses the power of convolution (CNN) and transformers to capture both small local features and larger global patterns in images. Unlike typical CNNs, which primarily focus on close pixels, MaxViT employs an attention mechanism that is especially useful for medical images with subtle texture and structural differences. At the same time, it retains convolutional processes to preserve essential low-level characteristics, such as edges, color changes, and lesion borders. The "Tiny" edition is smaller and more efficient, making it quicker and more practical while maintaining high performance.

2.4. Proposed hybrid model

Our research employs a parallel feature fusion strategy that combines two complementary neural networks operating in parallel as shown in Fig. 3. We selected two pre-trained models that bring different strengths, one is MobileNetV3 serves as our lightweight convolutional expert, excelling at capturing edges and fine-grained details. It is particularly good at identifying small but important features that might indicate specific conditions. The second model, MaxViT, uses a combination of window-based and grid-based attention mechanisms.

Before fusion, all input images are resized to 224 × 224 × 3. We remove the original classification heads from both networks and retain only their feature-extraction backbones. The architecture processes images through a parallel pathway. From each backbone, we extract the final pooled embedded features immediately before the original classifier. Specifically, MobileNetV3-Large produces a 960-dimensional feature vector (F_m), while MaxViT-Tiny produces a 512-dimensional feature vector (F_v). We then concatenate these vectors to form a single fused representation as shown in Eq. (1).

$$F = [F_m \parallel F_v] \in \mathbb{R}^{1472} \tag{1}$$

The fused feature vector is fed into a linear classification head consisting of a single fully connected layer, Linear($1472 \rightarrow C$) (where $C = 8$), which learns from both CNN- and Transformer-derived representations to produce the final predictions across the eight classes.

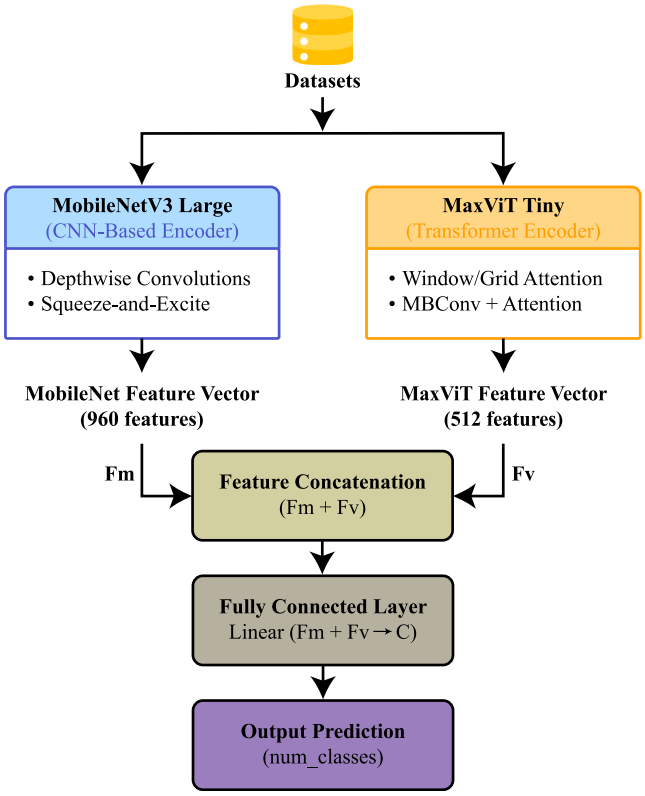


Fig. 3. Architecture of Proposed Hybrid Model.

2.5. Custom CNN model

We further extend our experiments by designing a custom CNN architecture that focuses on an ensemble strategy with VGG16 and ResNet101. The architecture of the custom CNN model, as shown in Fig. 4, is designed for effective image processing and classification.

The proposed CNN model is designed to learn meaningful patterns from images in an efficient and structured manner. Each block consists of a convolutional layer followed by a ReLU activation and a max-pooling operation, which helps the network progressively extract more informative features. ReLU activation functions enable learning of non-linear features. Max-pooling layers reduce the spatial size of the feature maps, improving computational efficiency and allowing the model to focus on the most relevant information. Dropout layers with rates of 20% and 30% are incorporated at different stages to reduce overfitting by randomly disabling a portion of neurons during training.

As shown in Fig. 4, the network takes an input image of size (224 × 224 × 3) and processes it through a sequence of convolutional blocks. The spatial resolution of the feature maps is gradually reduced (from (222 × 222) down to (5 × 5)), while the number of channels increases from 16 to 512. The final feature maps are flattened into a vector of size 12800 and passed to a fully connected layer with 800 neurons using ReLU activation. Finally, the output layer consists of 8 neurons with a softmax activation function, producing class probabilities for multi-class classification. This architecture achieves a balanced trade-off between feature richness, regularization, and computational efficiency. Additionally, layerwise summary of custom CNN model architecture as shown in Table 3.

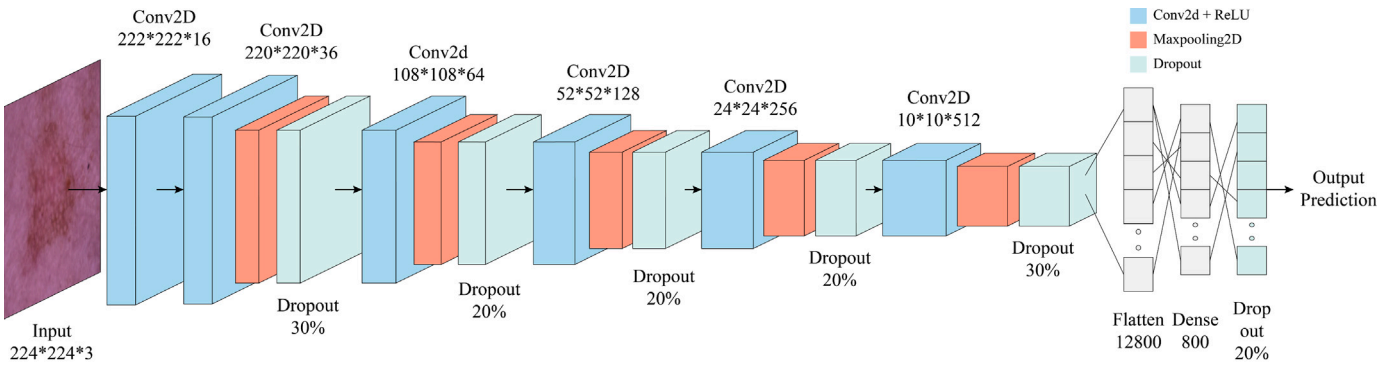


Fig. 4. Architecture of custom CNN model.

Table 3
Layer-wise summary of the custom CNN model.

Layer (type)	Output shape	Param #
conv2d (Conv2D)	(None, 222, 222, 16)	448
conv2d_1 (Conv2D)	(None, 220, 220, 36)	5220
max_pooling2d (MaxPooling2D)	(None, 110, 110, 36)	0
dropout (Dropout)	(None, 110, 110, 36)	0
conv2d_2 (Conv2D)	(None, 108, 108, 64)	20,800
max_pooling2d_1 (MaxPooling2D)	(None, 54, 54, 64)	0
dropout_1 (Dropout)	(None, 54, 54, 64)	0
conv2d_3 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 26, 26, 128)	0
dropout_2 (Dropout)	(None, 26, 26, 128)	0
conv2d_4 (Conv2D)	(None, 24, 24, 256)	295,168
max_pooling2d_3 (MaxPooling2D)	(None, 12, 12, 256)	0
dropout_3 (Dropout)	(None, 12, 12, 256)	0
conv2d_5 (Conv2D)	(None, 10, 10, 512)	1,180,160
max_pooling2d_4 (MaxPooling2D)	(None, 5, 5, 512)	0
dropout_4 (Dropout)	(None, 5, 5, 512)	0
flatten (Flatten)	(None, 12800)	0
dense (Dense)	(None, 800)	10,240,800
dropout_5 (Dropout)	(None, 800)	0
dense_1 (Dense)	(None, 8)	6408

Total params: 11,822,860 Trainable params: 11,822,860 Non-trainable params: 0.

2.6. Ensemble based approach

The capacity of ensemble approaches to improve prediction precision and adaptability has led to a significant growth in their use in deep learning. Ensemble approaches use the insights of individuals by combining the predictions of numerous models, instead of depending just on a single prediction model. In cases where individual models may be prone to overfitting or underfitting, this method can result in more reliable and accurate predictions.

We combined pre-trained VGG16, ResNet101, and our custom CNN model to build an ensemble model. This ensemble method was applied to the Skin Disease Dataset, which is relatively small, in order to improve learning stability and overall performance. This approach improves learning performance by bringing together the strengths of multiple models into a single, more powerful system. Each model contributes differently to the final prediction, allowing the ensemble to capture a wider range of visual features. We evaluated the performance of each individual model as well as several ensemble combinations to identify the most effective setup. To control the contribution of each model, weighted averaging was applied, and the weights were manually tuned as hyperparameters to achieve the best balance. The final ensemble model uses the optimal weights for each network, resulting in improved and more reliable classification performance.

3. Experimentation and results

This section presents a detailed experimental setup, including the training configuration, and a comprehensive discussion of the results obtained from the proposed models.

3.1. Experimental system setup

The experiment were conducted on a Linux machine (version 5.15.154+) equipped with a 64-bit ×86 CPU and 32 GB of RAM. This configuration offered ample processing power and memory to efficiently manage large datasets and demanding computations. For the deep learning tasks, a Tesla T4 GPU with 16 GB of memory was employed, coupled with CUDA 12.1, enabling faster model training through parallel processing. The models were developed and executed using PyTorch version 2.1.2, which further enhanced computational speed. To streamline data preprocessing, multi-core CPUs were leveraged, ensuring smooth and efficient data handling. The system also utilized Python 3.11, along with key libraries such as pandas 2.2.3, scikit-learn 1.2.2, and NumPy 1.26.4, to facilitate seamless data manipulation and machine learning workflows.

3.2. Training configuration

For optimization, our proposed hybrid model was trained using the AdamW optimizer with an initial learning rate of 1e−4. A StepLR learning rate scheduler was employed to reduce the learning rate by 90% after each epoch, while ensuring that it did not fall below a minimum threshold of 1e−6. The model was trained for a total of 20 epochs with a batch size of 32.

For the ensemble-based approach, the Adam optimizer was used with a learning rate of 1e−3. Early stopping was applied by monitoring the validation accuracy, and training was automatically halted when no further improvement was observed. The model was trained using batches of 128 for a maximum of 500 epochs. However, due to early stopping, training concluded after 50 epochs.

For both models, a model checkpoint mechanism was implemented to retain the best-performing weights based on validation performance, ensuring optimal model selection. Additionally, categorical cross-entropy was used as the loss function for all experiments.

3.3. Evaluation metrics

The following evaluation metrics have been used to evaluate the performance of the proposed skin disease detection and classification system.

3.3.1. Confusion matrix

The Confusion Matrix is a widely used matrix for model evaluation from which the accuracy, precision, recall, and F1-score can be derived. The model's ability to differentiate between various skin conditions and the quantitative measurements of the accuracy of its predictions are assessed by these metrics. A confusion matrix in multiclass classification extends the binary confusion matrix as shown in Table 4, to handle multiple classes, allowing for a more comprehensive comparison of actual and predicted labels. The matrix provides information on

Table 4
Confusion Matrix.

	Actual positive	Actual negative
Predicted Positive	True Positive (TP)	False Positive (FP)
Predicted Negative	False Negative (FN)	True Negative (TN)

the model's performance for each category, with each row indicating the true class and each column denoting the predicted class. In our skin disease classification with eight separate classes, the confusion matrix aids in evaluating not only overall accuracy (which can be misleading in imbalanced datasets), but also the precision, recall, and F1-score. False Positives (FP) measure how many actual negative class samples are predicted incorrectly, whereas True Positives (TP) measure how many actual positive class samples are predicted correctly. False Negatives (FN) occur when the model fails to predict the actual positive class samples, whereas True Negatives (TN) show how many actual negative class samples are predicted correctly. For example, if the model predicts BA-cellulitis" but the true label is "BA-impetigo", the result is an FP for "BA-cellulitis" and an FN for "BA-impetigo". These measures are crucial for comprehending the model's performance in predicting all classes of skin diseases as shown in Table 4.

3.3.2. Accuracy

Accuracy is a key parameter in classification tasks, expressed as the ratio of the sum of true positives (TP) and true negatives (TN) to the total number of samples, as shown in Eq. (2).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{FN} + \text{FP} + \text{TP} + \text{TN}} \tag{2}$$

3.3.3. Recall

Recall, which is determined by Eq. (3), quantifies the percentage of genuine positives among all actual positive cases to meet the limitation of accuracy. The model's ability to identify positive cases is demonstrated by a high recall, which is important to be maintained as missed positive diagnosis could have significant consequences. A higher FN reduces the recall since it represents missed positive testing.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

3.3.4. Precision

Precision is defined as the ratio of real positive predictions among all instances that were projected to be positive, as shown in Eq. (4). A high degree of precision indicates that the model is likely to be accurate in predicting positive instances, reducing the number of incorrect diagnoses and unnecessary treatments. A higher FP indicates inaccurate positive predictions reducing the Precision and the model's dependability,

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4}$$

3.3.5. F1-score

By combining precision and recall into a single statistic, the F1 Score is determined using Eq. (5) which offers an adequate evaluation providing a thorough understanding of the model's overall performance and is especially helpful in unbalanced situations.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

3.4. Simulation results

We report simulation results for individual pre-trained models as well as the custom CNN and the ensemble methods. Initially, individual pre-trained models (VGG16 and ResNet101) along with a custom CNN were implemented and evaluated on the Dataset 1. Subsequently, ensemble learning strategies were applied to further enhance classification performance. The highest accuracy of 97.86% is achieved using the ensembles of (CNN + ResNet101 + VGG16). We also evaluate the performance of the proposed hybrid architecture, which fusions the MobileNet and the MaxViT models. Table 5 presents the classification accuracy of the proposed hybrid technique, as well as the individual pre-trained deep learning models, the custom CNN, and the ensemble configurations on Dataset 1. Table 5 also summarizes the additional evaluation metrics, including accuracy, precision, recall, and F1- score, for both the proposed hybrid method and the ensemble model, providing a comprehensive comparison of their classification performance. Table 6 presents the classification accuracy on Dataset 2, comparing the proposed hybrid approach with the individual MobileNet and MaxViT models. As we can see from Tables 5 and 6 that the proposed hybrid model demonstrates outstanding and consistent performance across both datasets. For Dataset 1, the hybrid model achieves the highest accuracy of 99.57%, precision 99.59%, Recall 99.57%, and F1 score of 99.57%, while for Dataset 2, the hybrid model also shows the best performance with accuracy of 99.67%, precision 98.67%, Recall 98.67%, and F1 score of 98.67%.

The confusion matrices for the proposed hybrid network, the individual model and the ensemble method (ResNet101 + VGG16 + Custom CNN), are presented in Figs. 5(a)–5(d), respectively. Table 7 lists the eight skin disease class labels corresponding to the confusion matrix. Among the individual models, MaxViT demonstrates the strongest performance, as shown in Fig. 5(c), with only 5 misclassifications out of 234 test samples on Dataset 1. The proposed hybrid approach further improves classification accuracy, misclassifying only a single sample for Dataset 1, as illustrated in Fig. 5(a). In comparison, the ensemble model produces 5 incorrect predictions, as depicted in Fig. 5(d). For Dataset 2, the hybrid model misclassifies 336 samples out of 25,331 test images, as shown in Fig. 5(b).

In addition to these quantitative results, the accuracy and loss curves of the Proposed Hybrid Architecture using Datasets 1 and 2 are shown in Figs. 6 and 7. The training and validation accuracy and loss curves as illustrated in Figs. 6 and 7 show smooth convergence and close alignment, indicating stable learning behavior using both Dataset 1 and Dataset 2.

The class-wise accuracy, recall, and F1-score of the proposed hybrid technique for both datasets are presented in Table 8. It is observed that on Dataset 1, the model performs nearly perfectly and consistently across all disease categories. However, on Dataset 2, it produces good, well-balanced results across all eight types of lesions.

Consistent with these results, we further performed paired t-tests comparing the proposed hybrid model with its individual backbones (MobileNet and MaxViT) on both datasets. We perform a paired t-test by comparing the hybrid model and a backbone model with a confidence interval of 95%. From the results as included in Table 9, we can see that all p-values are below 0.05, which means that the hybrid model's performance is significantly better than both MobileNetV3 and MaxViT using both Dataset 1 and Dataset 2.

3.5. Model visualization interpretability

This section presents an interpretable analysis of the model's prediction behavior for skin disease image classification using the Grad-CAM++ [25] visualization method. The purpose of this approach is to reveal the image regions that most influence the model's classification decisions. Grad-CAM++ generates class-specific heatmaps, as illustrated in Fig. 8, which visually emphasize the areas within each

Table 5
Overall performance of all the implemented models on Dataset 1.

Model	Accuracy (%)	Precision	Recall	F1-Score
Custom CNN	82.48	83.00	82.00	82.00
ResNet101	96.58	97.00	97.00	97.00
VGG16	96.15	96.00	96.00	96.00
MobileNet	95.28	95.38	95.28	95.25
MaxViT	97.85	98.21	97.85	97.89
CNN+ResNet101+VGG16 (Ensemble)	97.86	97.98	97.86	97.86
Proposed Hybrid Architecture (MobileNet+ MaxViT)	99.57	99.59	99.57	99.57

Table 6
Overall performance of Proposed Hybrid Architecture and it corresponding individual models on Dataset 2.

Model	Accuracy (%)	Precision	Recall	F1-Score
MobileNet	91.47	91.39	91.47	91.41
MaxViT	97.89	97.89	97.89	97.89
Proposed Hybrid Architecture (MobileNet+ MaxViT)	98.67	98.67	98.67	98.67

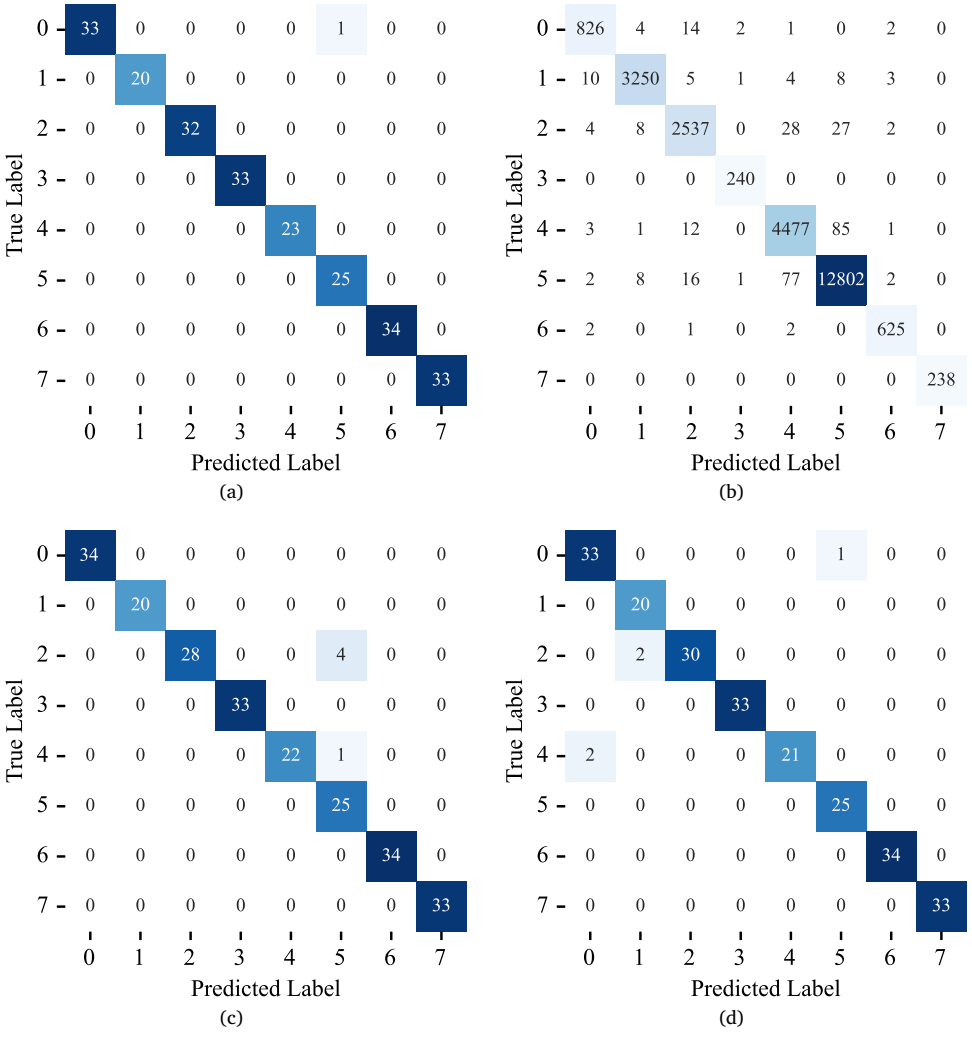


Fig. 5. Confusion matrices illustrating the classification performance of different models across datasets: (a) Proposed hybrid architecture on Dataset 1, (b) Proposed hybrid architecture on Dataset 2, (c) MaxViT architecture on Dataset 1, and (d) Ensemble model on Dataset 1.

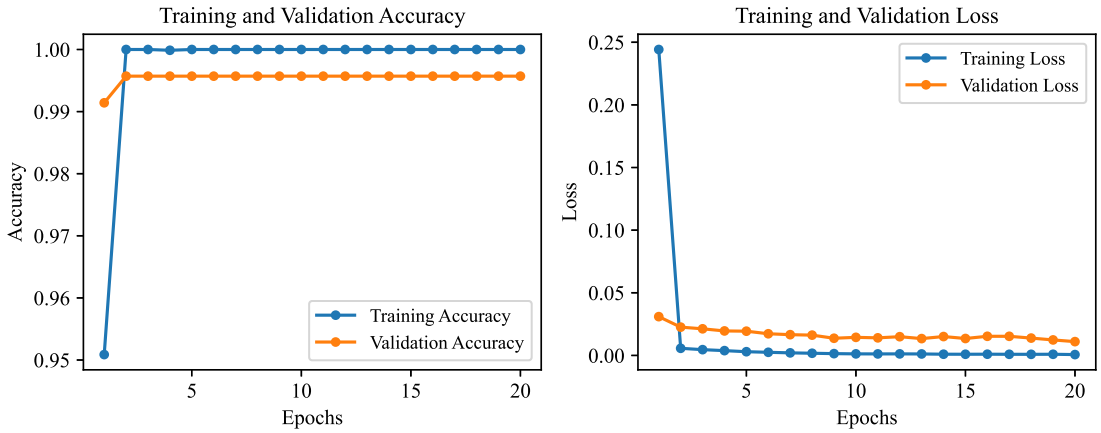


Fig. 6. Accuracy and loss curve of the proposed hybrid architecture for Dataset 1.

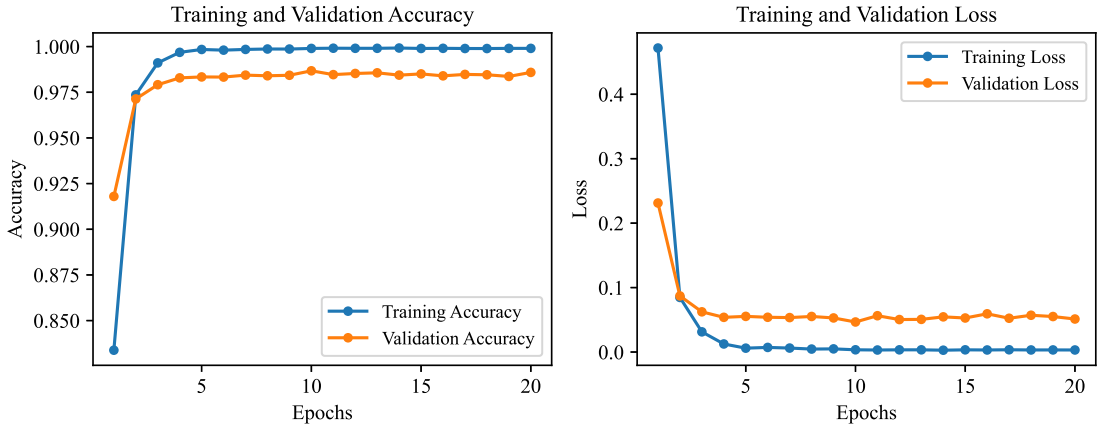


Fig. 7. Accuracy and loss curve of the proposed hybrid architecture for Dataset 2.

Table 7
Dataset's Classes and Labels.

Dataset 1	Dataset 2	Label
BA-Cellulitis	Actinic Keratosis (AK)	0
BA-Impetigo	Basal Cell Carcinoma (BCC)	1
FU-Athlete-foot	Benign Keratosis (BKL)	2
FU-Nail-fungus	Dermatofibroma (DF)	3
FU-Ringworm	Melanoma (MEL)	4
PA-Cutaneous-Larva-Migrans	Melanocytic Nevus (NV)	5
VI-Chickenpox	Squamous Cell Carcinoma (SCC)	6
VI-Shingles	Vascular Lesion (VASC)	7

image that contribute most strongly to the predicted outcome. The method operates by computing a weighted aggregation of feature maps from the final layer, where the weights are derived from the positive partial derivatives of the target class score. This process enables a clear visual interpretation of how the model associates image features with specific class labels.

4. Discussion

4.1. Optimum model

We compare the evaluation metrics as presented in Tables 5, 6, 8 and 9 to evaluate the overall performance of different models. The best model has been chosen based on the model's performance and the computational time as well as the complexity of the proposed models. The performance of the models are evaluated using evaluation metrics like Accuracy, Precision, Recall, and F1-score. The performance metrics for individual, ensemble models and the proposed hybrid models are presented in Tables 5 and 6. A higher F1 score indicates a better balance between false positives and false negatives as it is computed using the precision and recall scores. Therefore, a hybrid model with a smaller number of deep models achieving optimal accuracy and F1 score can be considered to qualify the model as the best model with less computational complexity. Considering the evaluation scores as presented in Tables 5 and 6, the hybrid model including the two deep learning models (MobileNet and MaxViT) may be chosen as the best model achieving the highest accuracy, precision, Recall, and F1 score for both Datasets 1 and 2. Considering the class-wise performance and runtime (in minutes) as presented in Table 8, the hybrid model may also be considered as the best choice for skin disease classification with less computational complexity as it achieves the highest accuracy of 99.57% and 98.67% using Datasets 1 and 2, respectively by combining the two deep learning models, MobileNetV3 and MaxViT. Additionally, based on the paired t-test results as presented in Table 9, it may also be concluded that the hybrid model's performance is significantly better than both MobileNetV3 and MaxViT using Datasets 1 and 2. Therefore, based on the four performance evaluation scores, computational complexity and the paired t-test, we may conclude that the hybrid model performs the best among all other individual and ensemble models by combining the two deep learning models, MobileNetV3 and MaxViT.

In the proposed Hybrid model, the MobileNetV3 is a lightweight CNN that can achieve high accuracy while utilizing less computational resources and memory, allowing it to learn meaningful visual information with fewer parameters. MobileNetV3 also incorporates attention-like modules (such as squeeze-and-excite) and enhanced activation algorithms to help focus on the most crucial aspects of the

Table 8
Class-wise precision, recall, F1-Score and accuracy of the proposed hybrid model for Dataset 1 and Dataset 2.

Dataset	Class	Precision	Recall	F1-Score	Accuracy (%)	Training time (min)
Dataset 1	BA-Cellulitis	1.00	0.97	0.98	99.57 ± 0.40	48.3
	BA-Impetigo	1.00	1.00	1.00		
	FU-Athlete-foot	1.00	1.00	1.00		
	FU-Nail-fungus	1.00	1.00	1.00		
	FU-Ringworm	1.00	1.00	1.00		
	PA-Cutaneous-Larva-Migrans	0.96	1.00	0.98		
	VI-Chickenpox	1.00	1.00	1.00		
	VI-Shingles	1.00	1.00	1.00		
Dataset 2	Actinic Keratosis (AK)	0.98	0.97	0.97	98.67 ± 0.03	632.3
	Basal Cell Carcinoma (BCC)	0.99	0.99	0.99		
	Benign Keratosis (BKL)	0.98	0.97	0.98		
	Dermatofibroma (DF)	0.98	1.00	0.99		
	Melanoma (MEL)	0.98	0.98	0.98		
	Melanocytic Nevus (NV)	0.99	0.99	0.99		
	Squamous Cell Carcinoma (SCC)	0.98	0.99	0.99		
	Vascular Lesion (VASC)	1.00	1.00	1.00		

Table 9
Paired t-test results: Hybrid model vs Individual models.

Dataset	Comparison	t-stat	p-value	Decision
Dataset 1	MobileNet	3.82	0.0066	Significant ($p < 0.05$)
	MaxViT	4.12	0.0045	Significant ($p < 0.05$)
Dataset 2	MobileNet	5.32	0.0011	Significant ($p < 0.05$)
	MaxViT	4.69	0.0022	Significant ($p < 0.05$)

skin disease images. However, a deeper model requires extensive GPU and time to do the same task. MaxVit-Tiny captures both small local features and larger global patterns in images and employs an attention mechanism that is especially useful for medical images with subtle texture and structural differences.

4.2. Comparison with other works in the literature

We compare our proposed approach of skin disease classification with other works in the state-of-the-art literature, as presented in Table 10. The data source, data size and diversity, proposed algorithm or methodologies, and performance metrics have been compared in Table 10. Our proposed hybrid model has been applied on two different Datasets, "Skin-Disease-Dataset" with a smaller dataset of only 1159 images and a high volume public dataset "ISIC 2019 Skin Lesion Images for Classification Dataset" of 25331 images to introduce the limitations that can affect real-world skin disease diversity and external generalization. In our proposed hybrid model, only two deep learning models have been combined to obtain an accurate and best performance in skin disease classification with less computational complexity. The proposed hybrid model follows parallel feature extraction and feature-level fusion. MobileNet (CNN encoder) is well-suited for local texture and fine-grained patterns, while MaxViT (Transformer-based encoder) captures global context and long-range relationships using attention and grid/window interactions. The lesion classes contain both local cues (texture, small color changes, edges) and global cues (overall lesion shape, symmetry, structure). Therefore, concatenating the two feature vectors yields a richer representation than either model alone.

As we can see in Table 10 that the proposed Hybrid model integrating the MobileNet and MaxViT demonstrates superior performance compared to other similar works in the literature. It is observed that our proposed hybrid model outperforms other works in the literature in terms of classification accuracy, precision, recall and F1-score using both Datasets, demonstrating the effectiveness of the proposed hybrid model considering the real-world challenges with skin disease diversity and external generalization.

5. Conclusion

In this paper, we have presented several methods of skin disease detection and classification using individual models, ensemble models and the proposed hybrid model. We evaluated the performance of the models using two different Datasets of eight classes of skin disease images using several Performance evaluation metrics. We have applied data preprocessing and image augmentation before applying it to the individual and the proposed hybrid model. The proposed hybrid model integrating only two deep learning models has achieved the best accuracy by combining the features of MobileNet and MaxVit deep learning models. Considering class-wise performance, computational complexity, GPU consumption and the overall evaluation scores, the proposed hybrid model (MobileNet+MaxVit) has been considered as the best model, achieving the best accuracy by combining only two deep learning models over both Datasets. The proposed hybrid model has achieved an accuracy of 99.67% and 98.67% using Dataset 1 "Skin-Disease-Dataset" and Dataset 2 "ISIC 2019 Skin Lesion Images for Classification", respectively. Our proposed hybrid model has also outperformed the literature in terms of four performance evaluation scores using both datasets, considering the practical challenges of skin disease diversity and external generalization. Therefore, it may be concluded that the proposed hybrid model will be helpful for the dermatologist in real practice for proper skin disease diagnosis and treatment. However, the reported performance using Datasets 1 and 2 may be impacted by the variations in dataset size, possible biases, variability in lighting, and device heterogeneity.

CRediT authorship contribution statement

Shakif Ahmed: Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – review & editing. **K.M. Sadman Sakib:** Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft. **Umma Hany:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 10
Comparison with other works in the literature.

Author/Year/Index	Dataset name	Dataset size	Algorithm	Evaluation metrics
He et al. (2021) [16]	Xiangya-Derm	150 233	SECNN	Precision: 71.2, Recall: 77.9, F1: 74.4
Elashiri et al. (2022) [21]	PH2, MNIST-HAM10000	–	HSBSO-LSTM	Accuracy: 93.8, Precision: 90.4, F1: 92
Inthiyaz et al. (2023) [18]	Xiangya-Derm	–	Softmax-Classifier	Accuracy: 87
Hao et al. (2023) [22]	ISIC-2019, Private acne images	25331 and 2900	ConvNeXt-ST-AFF	Accuracy: 92.16, Precision: 90.96, Recall: 87.08, F1: 88.83
Proposed model	Skin-Disease-Dataset	1159	Hybrid (MobileNet +MaxViT)	Accuracy: 99.57, Precision: 99.59, Recall: 99.57, F1: 99.57
	ISIC 2019 Skin Lesion Images for Classification	25 331		Accuracy: 98.67, Precision: 98.67, Recall: 98.67, F1: 98.67

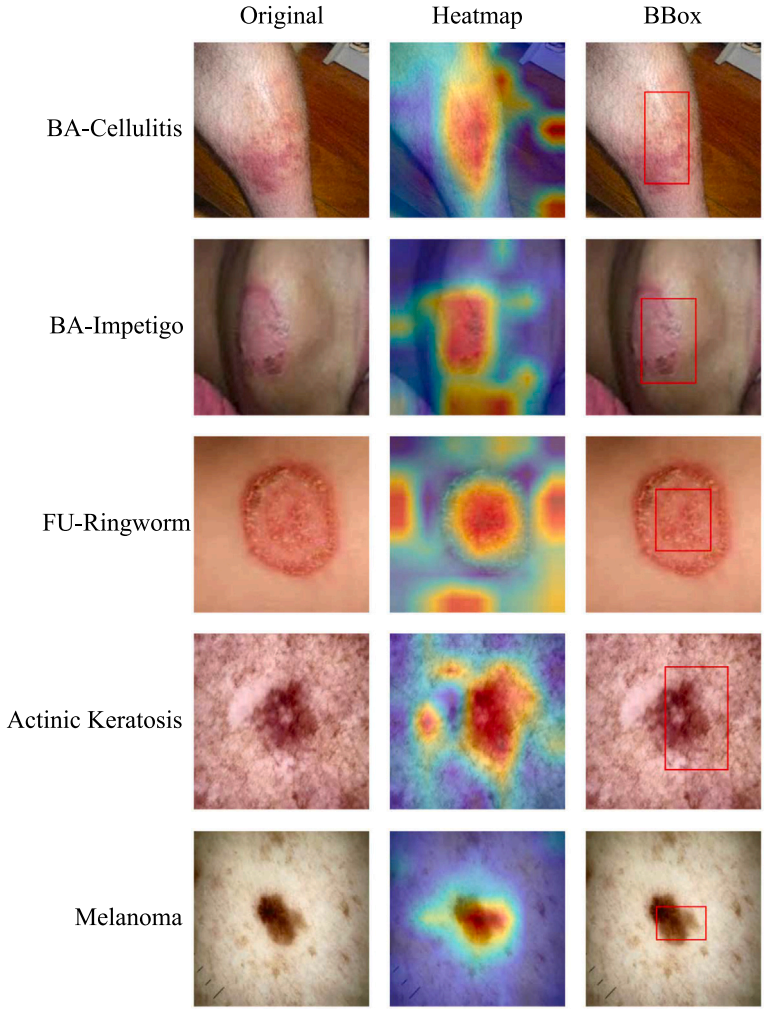


Fig. 8. Visual interpretability using Grad-CAM++ for representative samples from different skin disease classes.

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